Date: 14 / 10 / 2024

MAT - 2305 - Real Analysis II

Outlines

Infinite Series - Chapter 01
Function of several variables - Chapter 03
Riemann Integral - Chapter 02
Metric spaces (Introduction) - Chapter 04.

Assesment strategy

* Homework Assignment - 15%. 1 or 2 assignments (Assignment due dates will be communicated).

* Mid Exam - 25%. (Date: 02.12.2024) - (45 min)

* End Exam - 60% (3 hours) - 6 Questions.

Reference

1 Rudin W (1976) principles of Mathematical Analysis 3rd edition

2 A postal T.M. (1996), Mathematical Analysis 2nd edition.

Chapter 01 - Infinite series

$(x_n) = \langle x_n \rangle = \{x_n\} = \{x_n\}_{n=1}^{\infty}$ ← Sequence

$(x_n) = (x_1, x_2, x_3, \ldots)$

$\left(\frac{1}{n}\right) = \left(1, \frac{1}{2}, \frac{1}{3}, \ldots\right)$

$\left\{\frac{1}{n}\right\}_{n=1}^{100} = \left(1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{100}\right)$

Let (a_n) be a sequence of Real numbers we associate another sequence (S_n) with (a_n). Such that

$S_n = a_1 + a_2 + a_3 + \cdots + a_n$ for all $n \in \mathbb{N}$

$\mathbb{N} = \{1, 2, 3, 4, \ldots\}$

$\mathbb{N}$ = (The set of natural numbers)

Convergent $\longrightarrow \sum_{k=0}^{\infty} \frac{2}{(-3)^k} = \left\{ 2, \frac{4}{3}, \frac{14}{9}, \frac{40}{27}, \frac{122}{81}, \dots \right\}$

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The sequence (S_n) is called an infinite series. (or just a series) and it is denoted by the expression,

$$\sum_{n=1}^{\infty} a_n \equiv (S_n)$$

$$S_1 = a_1$$
$$S_2 = a_1 + a_2$$
$$S_3 = a_1 + a_2 + a_3$$
$$\vdots \qquad \vdots \qquad \vdots \qquad \vdots$$

- S_n is called the n^{th} partial sum of the infinite series (S_n)
- The infinite series $\sum_{n=1}^{\infty} a_n$ is said to be convergent (අභිසාරී) or be divergent (අපසාරී). (can't be both)
- In the sequence (S_n) converge to S, then S is called the sum of the infinite series and we write

$$\boxed{\sum_{n=1}^{\infty} a_n = S}$$

$$(S_n) = (S_1, S_2, S_3, \dots, S_n)$$

i.e $\sum_{n=1}^{\infty} a_n = \lim_{n \to \infty} S_n = \lim_{n \to \infty} \left[\sum_{k=1}^{n} a_k \right] = S$

Example - 01 Prove that $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$

Solution Let $a_n = \frac{1}{n(n+1)}$ for all $n \in \mathbb{N}$

$$a_n = \frac{1}{n} + \frac{-1}{(n+1)} = \frac{1}{n} - \frac{1}{(n+1)}$$

$$a_1 = \frac{1}{1} - \cancel{\frac{1}{2}} = \frac{1}{2}$$

$$a_2 = \cancel{\frac{1}{2}} - \cancel{\frac{1}{3}} = \frac{1}{6}$$

$$a_3 = \cancel{\frac{1}{3}} - \frac{1}{4} = \frac{1}{12}$$

$$\vdots \qquad \vdots \qquad \vdots \qquad \vdots$$

+

Divergent $\longrightarrow$ $\sum_{k=1} (-1)^k = -1+1-1+1-1+1-\cdots$

අපසාරී / [illegible]

[illegible]

$$S_n = \begin{cases} -1 & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

$$a_{n-1} = \cancel{\frac{1}{n-1}} - \cancel{\frac{1}{n}} = \frac{1}{n(n-1)}$$

$$a_n = \cancel{\frac{1}{n}} - \frac{1}{(n+1)} = \frac{1}{n(n+1)}$$

Adding all the n^{th} equations given.

$$\sum_{k=1}^{n} a_k = 1 - \frac{1}{(n+1)} = S_n$$

$$\sum_{n=1}^{\infty} a_n = \lim_{n \to \infty} \left[1 - \frac{1}{(n+1)}\right] = \underline{\underline{1}}$$

Example - 02. Consider $\sum_{n=1}^{\infty} 1$

Solution Let $a_n = 1$ for all $n \in \mathbb{N}$

Then, $S_n = 1+1+1+\cdots\cdots+1 = n$

$$\therefore \sum_{n=1}^{\infty} 1 = \lim_{n \to \infty} S_n = \lim_{n \to \infty} n = \infty$$

Example - 03 Consider $\sum_{n=1}^{\infty} 0$

Solution Let $a_n = 0$ for all $n \in \mathbb{N}$

Then, $S_n = 0+0+\cdots\cdots+0 = 0$

$$\therefore \sum_{n=1}^{\infty} 0 = \lim_{n \to \infty} S_n = \lim_{n \to \infty} 0 = 0$$

Example - 04 Consider $\sum_{n=1}^{\infty} \frac{1}{n}$ (Harmonic Series) [illegible]

Solution: Let $a_n = \frac{1}{n}$ for all $n \in \mathbb{N}$

Then, $S_n = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n}$

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Note

- A sequence (x_n) is called a Cauchy sequence if for every $\varepsilon > 0$ $\exists$ an integer $N \in \mathbb{N}$ such that $|x_n - x_m| < \varepsilon$ whenever $m, n \geq N$

Note that the sequence (S_n) is not Cauchy

Claim

Take any two positive integers m, n with $n > m$

Now consider.

$$|S_n - S_m| = \left| \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{m} + \frac{1}{m+1} + \cdots + \frac{1}{n}\right) - \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{m}\right) \right|$$

$$= \frac{1}{m+1} + \frac{1}{m+2} + \cdots + \frac{1}{n}$$

$$\geq \frac{1}{n} + \frac{1}{n} + \cdots + \frac{1}{n}$$

$$= (n-m)\frac{1}{n} = 1 - \frac{m}{n}$$

for all $m, n \in \mathbb{N}$ with $n > m$.

i.e $|S_n - S_m| \geq 1 - \frac{m}{n}$ for all $n \in \mathbb{N}$ with $n > m$

Let $n = 2m$ $(> m)$

Then $|S_{2m} - S_m| > 1 - \frac{m}{2m} = \frac{1}{2} > 0$

for all $m \in \mathbb{N}$.

$\therefore$ (S_n) is not cauchy.

Remark 1

A Sequence of real numbers is convergent iff it is Cauchy

Hense (S_n) is not cauchy $\Rightarrow$ (S_n) is not Convergent.

$\therefore$ (S_n) is divergent.

Also $S_{n+1} - S_n = \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} + \frac{1}{n+1}\right) - \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{n}\right)$

$= \frac{1}{n+1} > 0 \quad \forall\ n \in \mathbb{N}$

$\therefore$ (S_n) is monotonic increasing and it is not bounded above.

(S_n) is increasing and divergent

$\therefore \displaystyle\sum_{n=1}^{\infty} \frac{1}{n} = \infty$

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*** Theorem 1

If $\displaystyle\sum_{n=1}^{\infty} a_n$ is convergent, then $\displaystyle\lim_{n\to\infty} a_n = 0$

Proof

Let $S_n = \underbrace{a_1 + a_2 + a_3 + \dots + a_{n-1}}_{S_{n-1}} + a_n$

$\Rightarrow S_n = S_{n-1} + a_n$

$\Rightarrow a_n = S_n - S_{n-1}$

Since $\displaystyle\sum_{n=1}^{\infty} a_n$ is convergent, (S_n) is also convergent.

Let $\displaystyle\lim_{n\to\infty} S_n = S \in \mathbb{R}$

Now ① $\Rightarrow \displaystyle\lim_{n\to\infty} a_n = \lim_{n\to\infty} (S_n - S_{n-1})$

$= \displaystyle\lim_{n\to\infty} S_n - \lim_{n\to\infty} S_{n-1}$

$= S - S$

$= 0$

DNE - Does Not Exist.

$\Rightarrow$ implied

$\not\Rightarrow$ does not implied

Series is either convergent or divergent. not both.

Note

This theorem says that $\sum_{n=1}^{\infty} a_n$ is convergent $\Rightarrow \lim_{n\to\infty} a_n = 0$

Is the ~~theorem~~ converse of theorem true ?

$\lim_{n\to\infty} a_n \not\Rightarrow \sum_{n=1}^{\infty} a_n$ is

Counter example

Consider $\sum_{n=1}^{\infty} \frac{1}{n}$

Let $a_n = \frac{1}{n}$ for all $n \in \mathbb{N}$.

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{1}{n} = 0$

But $\sum_{n=1}^{\infty} \frac{1}{n}$ is <u>not</u> convergent (see eg 4)

$\sum_{n=1}^{\infty} 0 = 0$

$a_n = 0 \ \forall n \in \mathbb{N}$.

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} 0 = 0$

Test for Divergence

If $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent.

Example - 05 Consider $\sum_{n=1}^{\infty} n \sin(1/n)$

Let $a_n = n \sin(1/n) \ \forall n \in \mathbb{N}$.

Then $\lim_{n\to\infty} a_n = \lim_{n\to\infty} n \sin(1/n)$

$$= \lim_{n\to\infty} \frac{\sin(1/n)}{(1/n)} = 1 \neq 0$$

$\therefore$ The Series $\sum_{n=1}^{\infty} n \sin(1/n)$ is divergent.

Example - 06 $\sum_{n=1}^{\infty} (-1)^n$

Let $a_n = (-1)^n \ \forall n \in \mathbb{N}$

$$a = \begin{cases} +1 & \text{if } n \text{ is even} \\ -1 & \text{if } n \text{ is odd} \end{cases}$$

$\therefore \lim_{n\to\infty} a_n \neq 0$

$\therefore \sum_{n=1}^{\infty} (-1)^n$ is divergent

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Alternative method

$S_n = a_1 + a_2 + a_3 + \dots + a_n$

$= (-1) + (1) + (-1) + \dots + (-1)^n = \begin{cases} 0 & \text{if } n \text{ is even} \\ -1 & \text{if } n \text{ is odd} \end{cases}$

So, $\lim_{n \to \infty} S_n$ DNE

$\therefore (S_n)$ is divergent. Hence $\sum_{n=1}^{\infty} (-1)^n$ is divergent.

Example - 07 Consider the geometric series with common ratio r, $\sum_{n=1}^{\infty} r^{n-1}$

Let $a_n = r^{n-1} \quad \forall n \in \mathbb{N}$

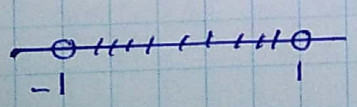
$S_n = a_1 + a_2 + a_3 + \dots + a_n$

$= 1 + r + r^2 + \dots + r^{n-1}$

$= \dfrac{1-r^n}{1-r}$ if $r \neq 1$

$\therefore S_n = \begin{cases} \dfrac{1-r^n}{1-r} & \text{if } r \neq 1 \\ n & \text{if } r = 1 \end{cases}$

Case 1 : $|r| < 1 \quad (-1 \leq r < 1)$



Then $\lim_{n \to \infty} S_n = \lim_{n \to \infty} \dfrac{1-r^n}{1-r} = \dfrac{1}{1-r}$

i.e the geometric series is convergent.

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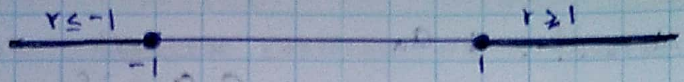
Case II : $|r| \geq 1$ ($r \geq 1$ or $r \leq -1$)



Let $a_n = r^{n-1}$

$\lim_{n \to \infty} a_n = \lim_{n \to \infty} r^{n-1} \neq 0$

$\therefore \sum_{n=1}^{\infty} a_n$ is not Convergent

So in a geometric series

1) $\sum_{n=1}^{\infty} r^{n-1} = \frac{1}{(1-r)}$ iff $|r| < 1$

2) $\sum_{n=1}^{\infty} r^{n-1}$ is divergent if $|r| \geq 1$

Theorem 2

a) If the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ Converge, then $\sum_{n=1}^{\infty} (a_n + b_n)$ converges and $\sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$

b) If the series $\sum_{n=1}^{\infty} a_n$ converges and c is a constant, then $\sum_{n=1}^{\infty} c \cdot a_n$ Converges and $\sum_{n=1}^{\infty} (c \cdot a_n) = c \sum_{n=1}^{\infty} a_n$.

Proof

a) Let S_n, t_n and U_n be the partial sums of $\sum_{n=1}^{\infty} a_n$, $\sum_{n=1}^{\infty} b_n$, and $\sum_{n=1}^{\infty} (a_n + b_n)$, respectively.

Then,

$$
\begin{aligned}
S_n &= (a_1 + b_1) + (a_2 + b_2) + \cdots + (a_n + b_n) \\
&= (a_1 + a_2 + \cdots + a_n) + (b_1 + b_2 + \cdots + b_n) \\
&= S_n + t_n
\end{aligned}
$$

since both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent, the sequence (S_n) and (t_n) are convergent

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The sequence $(s_n + t_n)$ is convergent.

$$\lim_{n\to\infty} (s_n + t_n) = \lim_{n\to\infty} s_n + \lim_{n\to\infty} t_n$$

$$= a + b, \text{ where}$$

$$a = \sum_{n=1}^{\infty} a_n \text{ and } b = \sum_{n=1}^{\infty} b_n$$

$$\Rightarrow \lim_{n\to\infty} U_n = a + b$$

$\therefore \sum_{n=1}^{\infty} (a_n + b_n)$ is convergent and $\sum_{n=1}^{\infty} (a_n + b_n)$

$$= \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$$

b)

Example - 08 Find $\sum_{n=1}^{\infty} \frac{2^n - 4^{n+2}}{5^{n-1}}$

Solution $\frac{2^n - 4^{n+2}}{5^{n-1}} = 2\left(\frac{2}{5}\right)^{n-1} - 4^3\left(\frac{4}{5}\right)^{n-1}$

Both $\sum_{n=1}^{\infty}\left(\frac{2}{5}\right)^{n-1}$ and $\sum_{n=1}^{\infty}\left(\frac{4}{5}\right)^{n-1}$ are geometric series with common ratios $r = \frac{2}{5}$ and $r = \frac{4}{5}$ respectively.

Therefore these series are convergent.

$$\boxed{\sum_{n=1}^{\infty} r^{n-1} = \frac{1}{r-1}, \ |r| < r}$$

Also, $\sum_{n=1}^{\infty}\left(\frac{2}{5}\right)^{n-1} = \frac{1}{1-\frac{2}{5}} = \frac{5}{3}$

and $\sum_{n=1}^{\infty}\left(\frac{4}{5}\right)^{n-1} = \frac{1}{1-\frac{4}{5}} = 5$

$$\sum_{n=1}^{\infty} \frac{2^n - 4^{n+2}}{5^{n-1}} = 2\sum_{n=1}^{\infty}\left(\frac{2}{5}\right)^{n-1} - 4^3\sum_{n=1}^{\infty}\left(\frac{4}{5}\right)^{n-1}$$

$$= 2 \cdot \frac{5}{3} - 64 \times 5$$

$$= -\frac{950}{3}$$

Theorem 3

Let $\sum_{n=1}^{\infty} a_n$ be an infinite series of non-negative terms. (i.e, $a_n \geq 0 \ \forall\, n \in \mathbb{N}$)

Then $\sum_{n=1}^{\infty} a_n$ converges if and only if the sequence (S_n) of its partial sums is bounded.

$\sum_{n=1}^{\infty} a_n$, with $a_n \geq 0$

$S_n = a_1 + a_2 + a_3 + \cdots + a_n$

$\sum_{n=1}^{\infty} a_n$ is cgt $\Leftrightarrow \underbrace{|S_n| \leq M < \infty}_{\text{bounded}}$

proof

Let $S_n = a_1 + a_2 + a_3 + \dots + a_n = \sum_{n=1}^{n} a_k$ for all $n \in \mathbb{N}$

Also, $S_{n+1} = a_1 + a_2 + a_3 + \dots + a_n + a_{n+1}$

$\Rightarrow S_{n+1} = S_n + a_{n+1} \quad \forall\, n \in \mathbb{N}$ —— (1)

Since, $a_{n+1} \geq 0 \quad \forall\, n \in \mathbb{N}$,

$S_{n+1} \geq S_n \quad \forall\, n \in \mathbb{N}$

So, the sequence (S_n) is monotonic increasing.

MCT (Monotonic Convergent Theorem)

A sequence (x_n) is cgt iff it is bounded. (x_n) is cgt $\Leftrightarrow |x_n| < \infty$

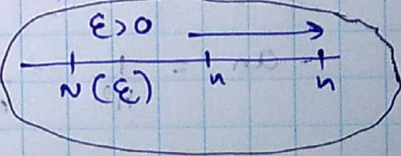
By monotonic convergence theorem (MCT) (S_n) is convergent iff (S_n) is bounded.

i.e $\sum_{n=1}^{\infty} a_n$ is convergent if and only if (S_n) is bounded.

Theorem 4

The infinite series $\sum_{n=1}^{\infty} a_n$ converges iff for each $\varepsilon > 0$ there exists a positive integers $N\varepsilon$, such that

$|a_{n+1} + a_{n+2} + \dots + a_m| < \varepsilon$ whenever $m > n > N(\varepsilon)$



proof

$\sum_{n=1}^{\infty} a_n$ converges iff (S_n) converges, where $S_n = a_1 + a_2 + a_3 + \dots + a_n$

iff (S_n) is Cauchy. iff for each $\varepsilon > 0$, there exists $N(\varepsilon) \in \mathbb{N}$ Such that

$|S_m - S_n| < \varepsilon$ whenever $m > n > N(\varepsilon)$

iff $|(a_1 + a_2 + \dots + a_m) - (a_1 + a_2 + \dots + a_n)| < \varepsilon$ whenever $m > n > N(\varepsilon)$

iff $|a_{n+1} + a_{n+2} + \dots + a_m| < \varepsilon$ whenever $m > n > N(\varepsilon)$

Theorem 5 (comparison test)

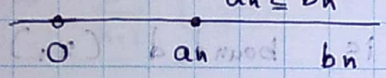
Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be infinite series of non-negative terms a_n, b_n with $a_n \leq b_n$ for all $n \in \mathbb{N}$.

Then,

a) If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges

b) If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ diverges

$a_n \leq b_n$



* $\sum_{n=1}^{\infty} b_n$ converges $\Longrightarrow \sum_{n=1}^{\infty} a_n$ converges

* $\sum_{n=1}^{\infty} a_n$ diverges $\Longrightarrow \sum_{n=1}^{\infty} b_n$ diverges

Example - 09 Show that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

Solution Let $a_n = \frac{1}{n^2} \quad \forall\, n \in \mathbb{N}$

$$a_n = \frac{1}{n^2} = \frac{1}{n.n} < \frac{1}{n(n-1)} \quad \text{for all } n = 2, 3, 4, \ldots$$

Let $b_n = \frac{1}{n(n-1)}$

$$\Longrightarrow b_n = -\frac{1}{n} + \frac{1}{n-1}, \quad n = 2, 3, 4, \ldots$$

$n = 2$,

$b_2 = -\cancel{\frac{1}{2}} + \frac{1}{1}$

$b_3 = -\cancel{\frac{1}{3}} + \cancel{\frac{1}{2}}$

$b_4 = -\frac{1}{4} + \cancel{\frac{1}{3}}$

$-\frac{1}{n} + \frac{1}{(n-1)}$

Adding all the $(n-1)$ eq^{ns} above gives

$$S_n = \sum_{k=2}^{n} b_k = 1 - \frac{1}{n} \longrightarrow \sum_{n=1}^{\infty} b_n = 1$$

$\therefore \sum_{n=2}^{\infty} b_n$ is cgt.

$$\frac{1}{n} < 1$$

Now, by the Comparison test (CT), $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is Convergent.

xample -10 Show that $\sum_{n=1}^{\infty} \frac{2^n}{3^n (n+1)^2 + 9}$ is Convergent.

olution Let $a_n = \frac{2^n}{3^n (n+1)^2 + 9} > 0 \quad \forall\, n \in \mathbb{N}$

$$a_n = \frac{2^n}{3^n (n+1)^2 + 9} < \frac{2^n}{3^n (n+1)^2} < \frac{2^n}{3^n} = \left(\frac{2}{3}\right)^n \quad \forall\, n \in \mathbb{N}$$

Let $b_n = \left(\frac{2}{3}\right)^n \quad \forall\, n \in \mathbb{N}$

$$= \frac{2}{3} \left(\frac{2}{3}\right)^{n-1}$$

Now, $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)^{n-1}$

$$= \frac{2}{3} \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^{n-1}$$

↑ Geometric serious with Common ratio $r_2 = \frac{2}{3}$

~~$r_2 = 2/3$~~ $\therefore$ It is convergent.

Now b_n CT, the series $\sum_{n=1}^{\infty} a_n$ is Convergent.

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Comparison Test

$a_n, b_n > 0$

Suppose $a_n \leq b_n \quad \forall n \in \mathbb{N}$

1. $\sum b_n$ cgts $\Rightarrow$ $\sum a_n$ cgs
2. $\sum a_n$ diverges $\Rightarrow$ $\sum b_n$ diverges.

Example-11. Consider $\sum_{n=1}^{\infty} \frac{1}{5n-3}$

Note that $\frac{1}{5n-3} > \frac{1}{5n} \quad \forall n \in \mathbb{N}$

Let, $a_n = \frac{1}{5n}$ and $b_n = \frac{1}{5n-3} \quad \forall n \in \mathbb{N}$

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{1}{5n} = \frac{1}{5} \sum_{n=1}^{\infty} \frac{1}{n} \text{ is divergent.}$$

Therefore, according to CT, $\sum_{n=1}^{\infty} b_n$ is divergent.

That is, $\sum_{n=1}^{\infty} \frac{1}{5n-3}$ is divergent.

Example-12 Consider $\sum_{n=1}^{\infty} \frac{1}{2n-15}$

Let $a_n = \frac{1}{2n-15}$ for all $n \in \mathbb{N}$

$$a_n \geq 0 \iff 2n - 15 \geq 0$$

$$\iff n \geq 7.5$$

$a_1, a_2, a_3, a_4, a_5, a_6, a_7 < 0$

That is $a_n \geq 0$ for all $n = 8, 9, 10, \ldots$

$$\sum_{n=1}^{\infty} a_n = (a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7) + \sum_{n=1}^{\infty} a_n$$

$$= \sum_{n=1}^{7} a_n + \sum_{n=1}^{\infty} a_n$$

now, $\frac{1}{2n-15} > \frac{1}{2n}$ for all $n = 8, 9, 10, \ldots \ldots (n \geq 8)$

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Since the Series $\sum_{n=1}^{\infty} \frac{1}{2n}$ is divergent. by CT, the series $\sum_{n=1}^{\infty} \frac{1}{2n+15}$ is divergent

Proof of CT (End Exame)

a) Suppose that $\sum_{n=1}^{\infty} b_n$ converges, where $b_n \geq 0$ for all $n \in \mathbb{N}$.

Let $b_n \geq a_n \geq 0$ for all $n \in \mathbb{N}$.

We must show that $\sum_{n=1}^{\infty} a_n$ Converges.

$\sum_{n=1}^{\infty} a_n$ Converges $\Leftrightarrow$ for each $\varepsilon > 0$ $\exists$ positive integer $N(\varepsilon)$.

Take any $\varepsilon > 0$

Then by the Cauchy criterian for series there exists $N(\varepsilon) \in \mathbb{N}$ Such that,

$$|b_{n+1} + b_{n+2} + \ldots\ldots + b_m| < \varepsilon \text{ ; whenever } m > n > N(\varepsilon) \quad — ①$$

Since $b_n \geq 0$ for all $n \in \mathbb{N}$,

$$\varepsilon > |b_{n+1} + b_{n+2} + \ldots\ldots + b_n| = \underbrace{b_{n+1}}_{\geq a_{n+1}} + \underbrace{b_{n+2}}_{\geq a_{n+2}} + \ldots + \underbrace{b_n}_{\geq a_n}$$

$$\geq a_{n+1} + a_{n+2} + \ldots\ldots + a_n$$

$$= |a_{n+1} + a_{n+2} + \ldots\ldots + a_n|$$

Since $a_n \geq 0$ $\forall n \in \mathbb{N}$

Now from ① we have

$$|a_{n+1} + a_{n+2} + \ldots\ldots + a_m| < \varepsilon \text{ whenever } m > n > N(\varepsilon)$$

Therefore, by Cauchy criterian for series, $\sum_{n=1}^{\infty} a_n$ Converges.

b) Suppose that $\sum_{n=1}^{\infty} a_n$ diverges.

Let, $S_n = \sum_{n=1}^{n} a_k$ for all $n \in \mathbb{N}$.

Since $a_n \geq 0$ for all $n \in \mathbb{N}$, by Theorem 3

the sequence (S_n) is unbounded.

Let $t_n = \sum_{k=1}^{n} b_k$ for all $n \in \mathbb{N}$

Since $b_n \geq a_n \geq 0 \; \forall n \in \mathbb{N}$.

$$a_1 + a_2 + a_3 + \cdots + a_n \leq b_1 + b_2 + \cdots + b_n$$

$$\sum_{k=1}^{n} a_k \leq \sum_{k=1}^{n} b_k$$

i.e. $S_n \leq t_n \; \forall n \in \mathbb{N}$.

Since (S_n) is unbounded, the sequence (t_n) is unbounded.

$\therefore \sum_{n=1}^{\infty} b_n$ is divergent. (Theorem 3)

Def^n

If $\sum_{n=1}^{\infty} |a_n|$ converges, then the series $\sum_{n=1}^{\infty} a_n$ is said to be absolutely convergent.

Example - 13 Consider $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$.

Let $a_n = \frac{(-1)^n}{n^2}$ for all $n \in \mathbb{N}$

$\Longrightarrow |a_n| = \left| \frac{(-1)^n}{n^2} \right| = \frac{1}{n^2}$ for all $n \in \mathbb{N}$

We know that $\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

$\therefore \sum_{n=1}^{\infty} \frac{(-1)^n}{A^2}$ is absolutely convergent.

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Theorem 6

If a series is absolutely convergent, then it is convergent.

Proof (MID EXAM)

Let, $\sum_{n=1}^{\infty} a_n$ be an absolutely convergent series.

We must show that $\sum_{n=1}^{\infty} a_n$ is absolutely convergent

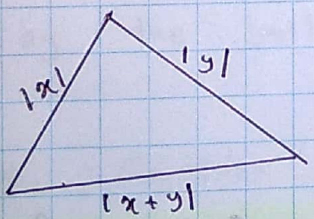
Since the series is absolutely convergent, the series $\sum_{n=1}^{\infty} |a_n|$ is convergent.

Then by the Cauchy criterion for each $\varepsilon > 0$, there exists $N(\varepsilon) \in \mathbb{N}$ such that

$$\Big|\underbrace{|a_{n+1}|}_{\geq 0} + \underbrace{|a_{n+2}|}_{\geq 0} + \cdots + \underbrace{|a_n|}_{\geq 0}\Big| < \varepsilon \quad \text{whenever } m, n > N(\varepsilon).$$

$$\Rightarrow |a_{n+1}| + |a_{n+2}| + \cdots + |a_n| < \varepsilon \quad \text{whenever } m, n > N(\varepsilon)$$
(since $|a_n| \geq 0$)

*Tri-angle Inequality



$$|x+y| \leq |x| + |y| \quad \forall\ x, y \in \mathbb{R}.$$

$$|x_1 + x_2 + \cdots + x_n| \leq |x_1| + |x_2| + \cdots + |x_n|$$

Now by the tri-angle inequality, we have

$$|a_{n+1} + a_{n+2} + \cdots + a_m| \leq |a_{n+1}| + |a_{n+2}| + \cdots + |a_n| \quad \text{—② }$$

Now, by ① and ②, we have

$$|a_{n+1} + a_{n+2} + \cdots + a_m| < \varepsilon \quad \text{whenever } m, n > N(\varepsilon)$$

i.e for each $\varepsilon > 0$, $\exists\ N(\varepsilon) \in \mathbb{N}$

s.t. $|a_{n+1} + a_{n+2} + \cdots + a_m| < \varepsilon$ whenever $m > n > N(\varepsilon)$

Now by the Cauchy criterion, $\sum_{n=1}^{\infty} a_n$ is convergent.

Example - 14 Consider $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

We will later show that $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is convergent (AST)

Let $a_n = \frac{(-1)^n}{n}$ for all $n \in \mathbb{N}$

$|a_n| = \left|\frac{(-1)^n}{n}\right| = \frac{1}{n}$ for all $n \in \mathbb{N}$.

Also we know that the series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

i.e., the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is convergent, but not it is absolutely convergent.

Note

Absolutely Convergent $\Longrightarrow$ Convergent

Convergent $\not\Longrightarrow$ Absolutely Convergent.

Theorem 7 (Limit form of CT)

Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be infinite series of positive terms a_n, b_n such that $\lim_{n\to\infty} \frac{a_n}{b_n} = l$

If $0 < l < \infty$, then the two series converge or diverge together.

$l = \lim_{n\to\infty} b_n \overset{\text{aus}}{\Longrightarrow} 0$

l

Example - 15. Consider $\sum_{n=1}^{\infty} \frac{n}{8n^3+9}$

$\sum \frac{1}{n^p}$ $p>1$

$0 \leq p \leq 1$

Let $a_n = \frac{n}{8n^3+9}$ for all $n \in \mathbb{N}$

Let $b_n = \frac{1}{n^2} > 0$ for all $n \in \mathbb{N}$

Now $l = \lim_{n\to\infty} \frac{a_n}{b_n}$

$= \lim_{n\to\infty} \left[\frac{\frac{n}{8n^3+9}}{1/n^2} \right]$

$= \lim_{n\to\infty} \left[\frac{n^3}{8n^3+9} \right]$

$= \lim_{n\to\infty} \left[\frac{1}{8 + 9/n^3} \right] = \frac{1}{8} > \infty$

we know that $\sum_{n=1}^{\infty} b_n$ is Convergent

so, by the limit form of CT, $\sum_{n=1}^{\infty} \frac{n}{8n^3+9}$ is cgt.

Example - 16. Consider $\sum_{n=1}^{\infty} \frac{n}{8n^3-9}$

Let $a_n = \frac{n}{8n^3-9}$

$a_1 = \frac{1}{8-9} = -1 < 0$

$a_2 = \frac{2}{8(2)^3-9} > 0$

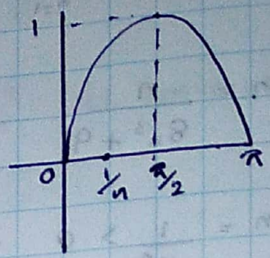
$\sum_{n=1}^{\infty} a_n = a_1 + \sum_{n=2}^{\infty} \underset{>0}{a_n}$

Let $b_n = \frac{1}{n^2} > 0 \quad \forall\, n \in \mathbb{N}$

Now, $l = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \left[\frac{\frac{n}{8n^3-9}}{1/n^2} \right] = \frac{1}{8} > 0$

Since $\sum b_n$ is cgt, by CT $\sum_{n=1}^{\infty} \frac{n}{8n^3-9}$ is cgt.

Example-17 Consider $\sum_{n=1}^{\infty} \sin(1/n)$



Let $a_n = \sin(1/n) \quad \forall \; n \in \mathbb{N}$

Let $b_n = \frac{1}{n} > 0 \quad \forall \; n \in \mathbb{N}$

$$l = \lim_{n \to \infty} \frac{a_n}{b_n}$$

$$= \lim_{n \to \infty} \frac{\sin(1/n)}{(1/n)}$$

$$= \lim_{1/n \to \infty} \frac{\sin(1/n)}{1/n}$$

$$= 1 > 0$$

WKT - we know that

cts - continous

Also, WKT $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

Thus, by LFCT, the series $\sum_{n \geq 1}^{\infty} \sin(1/n)$ is divergent.

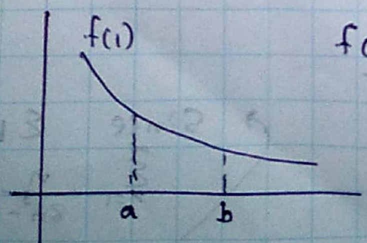
Theorem 8 (Intergral test)

Let f be a non-negative decreasing continuous function. defined on $[1, \infty]$.

Then, the series $\sum_{n \geq 1}^{\infty} f_n$ converges of and only if $\lim_{n \to \infty} [\int f(x)$ exists as a real number.

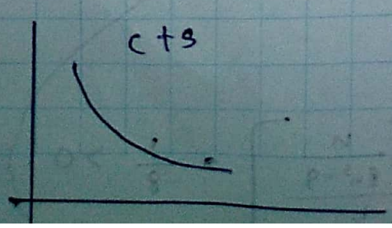
* $f(x) \geq \quad \forall \; x \in [1, \infty]$

* $f'(x) < 0 \quad \forall \; x \in [1, \infty)$



$f(a) \geq f(b)$ for $a < b$

*



$$\lim_{n \to \infty} \left[\int_1^n f(x)\, dx \right] \in \mathbb{R}$$

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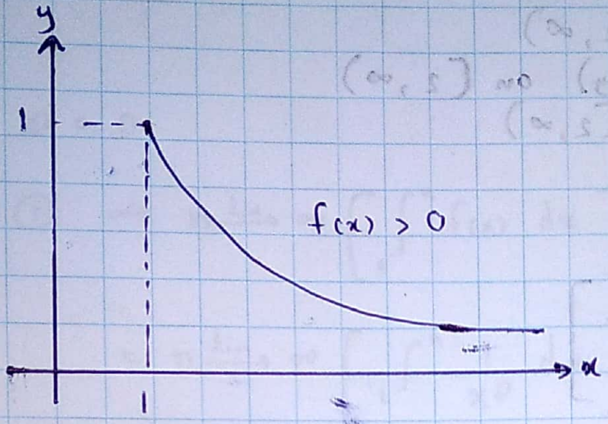
Example - 18 Use the integral test to show that $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

Here, $f_n = \frac{1}{n} \quad \forall n \in \mathbb{N}$

$f(x) = \frac{1}{x} \quad \forall x \in [1, \infty)$



Note

* $f(x) > 0 \quad \forall x \in [1, \infty)$
* $f(x)$ is decreasing on $[1, \infty)$
* $f(x)$ is cts on $[1, \infty)$.

So, we may apply the IT

Now consider,

$$\lim_{n \to \infty} \left[\int_1^n f(x)\, dx \right] = \lim_{n \to \infty} \left[\int_1^n \frac{1}{x}\, dx \right]$$

$$= \lim_{n \to \infty} \left[\ln|x| \Big|_{x=1}^{n} \right]$$

$$= \lim_{n \to \infty} \left[\ln(n) - \ln(1) \right]$$

$$= \infty \notin \mathbb{R}.$$

Now, by the IT the given series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

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Example - 19 Consider $\sum_{n=2}^{\infty} \frac{1}{n \ln(n)}$

Let $f(x) = \frac{1}{x \ln(x)} \quad \forall \; x \in [2, \infty)$

* $f(x) > 0 \quad \forall \; x \in [2, \infty)$
* $f(x)$ is $\downarrow$ (decreasing) on $[2, \infty)$
* $f(x)$ is cts on $[2, \infty)$

Use IT

$$\lim_{n \to \infty} \left[\int_2^n \frac{1}{x \ln(x)} \, dx \right]$$

$$= \lim_{n \to \infty} \left[\int_2^{\infty} \frac{1}{\ln(x)} \, d(\ln(x)) \right]$$

$$= \lim_{n \to \infty} \left[\ln(\ln(x)) \Big|_{x=2}^{n} \right]$$

$$= \lim_{n \to \infty} \left[\ln(\ln(n)) - \ln(\ln(2)) \right] = \infty$$

Hence, according to IT, the given series is not convergent.

Example - 20 Consider $\sum_{n=1}^{\infty} \frac{1}{n^p}$, where $p \in \mathbb{R}$. p - Series

We can discuss the convergence of the above series according to the above p.

Case I: $p < 0$

Then $\lim_{n \to \infty} \frac{1}{n^p} = \infty \neq 0$

Hence, $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is divergent.

Let $f(x) = \frac{1}{x^p}$ for $x > 0$

It is clear that f is a non-negative decreasing continuous function of $[1, \infty)$

Use IT

$$\lim_{n\to\infty} \int_1^n f(x)\,dx = \lim_{n\to\infty} \left[\int_1^n \frac{1}{x^p}\,dx \right] \quad \text{--- (1)}$$

$* \; p = 0$ $\qquad * \; p = 1$

$* \; 0 < p < 1$ $\qquad * \; p > 1$

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Case II : $p = 0$

Then (1) $\Rightarrow \lim_{n\to\infty} \left[\int_1^n f(x)\,dx \right]$

$$= \lim_{n\to\infty} \left[\int_1^n \frac{1}{x^p}\,dx \right]$$

$$= \lim_{n\to\infty} \left[\int_1^n 1\,dx \right]$$

$$= \lim_{n\to\infty} (n-1)$$

$$= \infty \notin \mathbb{R}$$

$\therefore \sum_{n=1}^{\infty} \frac{1}{n^p}$ is divergent.

Case III : $0 < p < 1$

$$\lim_{n\to\infty} \left[\int_1^n f(x)\,dx \right]$$

$$= \lim_{n\to\infty} \left[\int_1^n \frac{1}{x^p}\,dx \right]$$

$$= \lim_{n\to\infty} \left[\frac{x^{-p+1}}{-p+1} \Big|_{x=1}^{n} \right], \quad (0 < p < 1)$$

$$= \lim_{n\to\infty} \left[\frac{n^{-p+1}}{1} - \frac{1}{-p+1} \right]$$

$$= \lim_{n\to\infty} n^{-p} = \lim_{n\to\infty} \frac{1}{n^p}$$

$$= \lim_{n\to\infty} \frac{1}{n^{1/2}}$$

$$= \lim_{n\to\infty} n^{-p+1} = \lim_{n\to\infty} n^{-p}$$

$$= \infty$$

Case IV : $p = 1$

$$\lim_{n \to \infty} \left[\int_1^n \frac{1}{x^p}\, dx \right] = \lim_{n \to \infty} \left[\int_1^n \frac{1}{x}\, dx \right]$$

$$= \lim_{n \to \infty} \left[\ln|n| - 0 \right]$$

$$= \infty$$

$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^p}$ is divergent

Case V : $p > 1$

$$\lim_{n \to \infty} \left[\int_1^n \frac{1}{x^p}\, dx \right] = \lim_{n \to \infty} \left[\frac{x^{-p+1}}{-p+1} \Big|_{x=1}^{n} \right]$$

$$= \lim_{n \to \infty} \frac{1}{1-p} \left[\frac{1}{n^{p-1}} - 1 \right] = \frac{1}{p-1} \in \mathbb{R}$$ (since $p > 1$)

$\therefore \sum_{n=1}^{\infty} \frac{1}{n^p}$ is cgt.

$\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent is $p > 1$ and divergent if $p \leq 1$

Example - 21 Consider $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ is a p - series with $p = \frac{1}{2} < 1$.

Therefore, this series is divergent.

Example - 22 Consider $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$

This is a p - series with $p = \frac{3}{2} > 1$

Hence it is cgt.

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Example - 23 Consider $\sum_{n=1}^{\infty} \frac{(-1)^n (n^2+3)}{n^6 + 5n^3 + 7}$

Let $a_n = \frac{(-1)^n (n^2+3)}{n^6 + 5n^3 + 7} \quad \forall \; n \in \mathbb{N}$

$$|a_n| = \left| \frac{(-1)^n (n^2+3)}{n^6 + 5n^3 + 7} \right|$$

$$= \frac{n^2 + 3}{n^6 + 5n^3 + 7} \quad \forall \; n \in \mathbb{N}$$

$$\leq \frac{n^2 + 3n^2}{n^6} = \frac{4}{n^4}$$

$$\sum_{n=1}^{\infty} \frac{4}{n^4} = 4 \underbrace{\sum_{n=1}^{\infty} \frac{1}{n^4}}_{p\text{-series with } p = 4 > 1}$$

Since $\sum_{n=1}^{\infty} \frac{1}{n^4}$ is a p- series with $p = 4 > 1$. It is cgt

$\therefore$ By CT , the series $\sum_{n=1}^{\infty} |a_n|$ is cgt.

That is $\sum_{n=1}^{\infty} a_n$ is absolution convergent.

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent.

H/W Test the Convergence of $\sum_{n=2}^{\infty} \frac{1}{n \ln(n) \ln[\ln(n)]}$

Theorem 09 (Ratio test)

Let $\sum_{n=1}^{\infty} a_n$ be a series with non - zero terms.

a) If there exist a positive number $r < 1$ and a positive integer $N \in \mathbb{N}$. Such that,

$$\left|\frac{a_{n+1}}{a_n}\right| \leq r \quad \text{for} \quad n \geq N,$$

then $\sum_{n=1}^{\infty} a_n$ is absolutely convergent

b) If there exist a numbr $r \geq 1$ a positive integer $N \in \mathbb{N}$ Such that,

$$\left|\frac{a_{n+1}}{a_n}\right| \geq r \quad \text{for} \quad n \geq N$$

then $\sum a_n$ is divergent.

Theorem 10 (Limit form of ratio test)

Let $\sum_{n=1}^{\infty} a_n$ be a series with non - zero terms.

Let $\lim_{n \to \infty} \left|\frac{a_{n+1}}{a_n}\right| = l$

a) If $l < 1$, then $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.

b) If $l > 1$, then $\sum_{n=1}^{\infty} a_n$ is divergent

c) If $l = 1$, then the test is inconclusive.

Example - 24 Consider $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} 3^n}{n!}$

Let $a_n = \frac{(-1)^{n-1} 3^n}{n!} \quad \forall\, n \in \mathbb{N}$

Note $a_n \neq 0 \quad \forall \quad n \in \mathbb{N}$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(-1)^n 3^{n+1}}{(n+1)!} \times \frac{n!}{(-1)^{n-1} 3^n}\right|$$

$$= \left|\frac{-3}{n+1}\right| = \frac{3}{n+1}$$

$$l = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{3}{n+1} = 0 < 1$$

By LFRT, $\sum_{n=1}^{\infty} a_n$ is absolutely convergent and hence this series is convergent.

Example - 25 Consider $\sum_{n=1}^{\infty} \frac{1.3.5 \ldots (2n-1)}{n!}$

Let $a_n = \frac{1.3.5 \ldots (2n-1)}{n!} \quad \forall \quad n \in \mathbb{N}$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{1.3.5 \ldots (2n-1)(2n+1)}{(n+1)!} \times \frac{n!}{1.3.5 \ldots (2n-1)}\right| \quad (n+1)! = (n!)(n+1)$$

$$= \left|\frac{2n+1}{n+1}\right| = \frac{2n+1}{n+1} \quad \forall\, n \in \mathbb{N}$$

$$l = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|$$

$$= \lim_{n\to\infty}\frac{2n+1}{n+1} = \lim_{n\to\infty}\frac{2+1/n}{1+1/n} = 2 > 1$$

By LFRT, $\sum_{n=1}^{\infty} a_n$ is divergent.

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i) $\sum_{n=1}^{\infty} (-1)^n$ — divergent

ii) $\sum_{n=1}^{\infty} \frac{1}{n^2}$ — cgt

} use LFRT

$a_n = (-1)^n$

$a_{n+1} = (-1)^{n+1}$

$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(-1)^{n+1}}{(-1)^n}\right| = 1$

$\therefore\ l = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = 1$

Proof of Ratio Test

a) Let $\left|\frac{a_{n+1}}{a_n}\right| \le r < 1$ for all $n \ge N$, — ① where $a_n \ne 0\ \forall\ n \in \mathbb{N}$.

We must show that $\sum a_n$ is Convergent.

① $\Rightarrow |a_{n+1}| \le r|a_n|\ \forall\ n \ge N$

$n = N \rightarrow |a_{N+1}| \le r|a_N|$

$n = N+1 \Rightarrow |a_{N+2}| \le r|a_{N+1}| \Rightarrow |a_{N+2}| \le r^2|a_N|$

$n = N+2 \Rightarrow |a_{N+3}| \le r|a_{N+2}| \Rightarrow |a_{N+3}| \le r^3|a_N|$

So we have $|a_{N+j}| \le r^j |a_N|$ for $j = 1, 2, 3, \ldots$

Now, $\sum_{n=1}^{\infty} r^j |a_N| = |a_N| \underbrace{\sum_{j=1}^{\infty} r \cdot r^{j-1}}$

Geometric series with Comman ratio $r<1$

$\therefore$ This geometric series is cgt.

Now, by the CT the series $\sum_{j=1}^{\infty} |a_{N+j}|$ is convergent.

Therefore, $\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{N} |a_n| + \sum_{j=1}^{\infty} |a_{N+j}|$ is cgt.
(Absolutely cgt)

$\therefore \sum a_n$ is convergent.

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b) Suppose that $\left|\frac{a_{n+1}}{a_n}\right| \geq r > 1$ for all $n \geq N$

$\Rightarrow \underbrace{|a_{n+1}| > r|a_n| > |a_n|}$ for $n \geq N$ (since $r>1$)

$\Rightarrow |a_n| \leq |a_{n+1}|$; $n = N, N+1, N+2, \ldots$

$\therefore$

$|a_N| \leq |a_{N+1}| \leq |a_{N+3}| \ldots$

$(|a_{N+j}|)$; $j = 0, 1, 2, 3, \ldots$ is an increasing sequence.

$\rightarrow \lim_{n\to\infty} |a_n| \neq 0 \implies \lim_{n\to\infty} a_n \neq 0$

$\rightarrow$ The series $\sum a_n$ is divergent

Proof of LFRT is Left

Theorem 11 (Alternative Series Test - AST)

If (c_n) is a decreasing sequence of positive numbers with $\lim_{n\to\infty} c_n = 0$, then $\sum_{n=1}^{\infty} (-1)^{n-1} c_n$ and $\sum_{n=1}^{\infty} (-1)^n c_n$ Converge.

Example - 26 Consider $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

$c_n > 0$
c_n is $\downarrow$
$\lim_{n\to\infty} c_n = 0$
$\Bigg\}$ $\sum_{n=1}^{\infty} (-1)^{n-1} c_n$ and $\sum_{n=1}^{\infty} (-1)^n c_n$ - Cgt.

Let $c_n = \frac{1}{n}$ $\forall n \in \mathbb{N}$

i) $c_n > 0$ $\forall n \in \mathbb{N}$

$$c_n - c_{n+1} = \frac{1}{n} - \frac{1}{n+1}$$

$$= \frac{n+1-n}{n(n+1)} = \frac{1}{n(n+1)} > 0 \quad \forall n \in \mathbb{N}$$

$c_n - c_{n+1} > 0$ $\forall n \in \mathbb{N}$

$\therefore c_n > c_{n+1}$ $\forall n \in \mathbb{N}$

That is (c_n) is decreasing.

$$\lim_{n\to\infty} c_n = \lim_{n\to\infty} \frac{1}{n} = 0$$

There fore, by AST the given series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is Cgt.

Root Test

Definition Let (x_n) be a bounded Sequence of real numbers.

Then, $\limsup x_n$ and $\liminf x_n$ are defined as

$$\limsup x_n = \inf_{n\geq 1} \left(\sup_{k\geq n} x_k \right) \text{ and}$$

$$\liminf x_n = \sup_{n\geq 1} \left(\inf_{k\geq n} x_k \right), \text{ respectively.}$$

Example - 27 Let $((-1)^n)$

Let $((-1)^n)$

Here $x_n = (-1)^n \quad \forall\ n \in \mathbb{N}$

$(x_n) = (-1, 1, -1, 1, \ldots\ldots)$

Let $M_n = \text{Sup}\{x_n, x_{n+1}, x_{n+2}, \ldots\ldots\}$

$M_1 = \text{Sup}\{x_1, x_2, x_3, \ldots\ldots\}$

$= \text{Sup}\{-1, 1, -1, \ldots\ldots\} = 1$

$M_2 = \text{Sup}\{x_2, x_3, x_4, \ldots\}$

$= \text{Sup}\{1, -1, +1, \ldots\} = 1$

$M_3 = \text{Sup}\{x_3, x_4, \ldots\ldots\}$

$= \text{Sup}\{-1, 1, -1, \ldots\ldots\} = +1$

$\limsup x_n = \inf\{M_1, M_2, M_3, \ldots\}$

$= \inf\{1, 1, 1, \ldots\ldots\} = 1$

Let $m_n = \inf\{x_n, x_{n+1}, x_{n+2}, \ldots\}$

$m_1 = \inf\{-1, -1, -1, \ldots\ldots\} = -1$

$m_2 = -1$

$m_3 = -1$

$\liminf x_n = \text{Sup}\{m_1, m_2, m_3, \ldots\}$

$= \text{Sup}\{-1, -1, -1, \ldots\ldots\}$

$= -1$

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Consider $x_n = \frac{1}{n}$, $n = 1, 2, 3, \ldots$

$z_n = \sup\{x_n, x_{n+1}, x_{n+2}, \ldots\}$

$= \sup\{\frac{1}{n}, \frac{1}{n+1}, \frac{1}{n+2} \ldots\}$

$z_1 = \sup\{\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \ldots\} = 1$

$z_2 = \sup\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4} \ldots\} = 1/2$

$z_3 = \sup\{\frac{1}{3}, \frac{1}{4}, \ldots\} = 1/3$

Note that (z_n) is a decreasing sequence.

$\limsup = \inf\{z_1, z_2, z_3, \ldots\}$

$= \inf\{1, 1/2, 1/3, \ldots\}$

$= 0 \left(= \lim_{n\to\infty} x_n = \lim_{n\to\infty} 1/n\right)$

Let $z_n' = \inf\{x_n, x_{n+1}, \ldots\}$

$= \inf\{1/n, 1/n+1, 1/n+2, \ldots\}$

$z_1' = \inf\{1/1, 1/2, 1/3, \ldots\} = 0$

$z_2' = \inf\{1/2, 1/3, 1/4, \ldots\} = 0$

$z_3' = \inf\{1/3, 1/4, \ldots\} = 0$

Note that in general (z_n') is an increasing sequence, when (x_n) is bounded.

$\liminf x_n = \sup\{z_1', z_2', z_3', \ldots\}$

$= \sup\{0, 0, 0, \ldots\}$

$= 0 \left(= \lim_{n\to\infty} x_n = \lim_{n\to\infty} \frac{1}{n}\right)$

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Note

Let (x_n) be a bounded Sequence of real numbers.

Let $y_n = \sup\{x_n, x_{n+1}, x_{n+2}, \dots\}$

$= \sup_{k \geq n} x_k$

$\limsup x_n = \inf_{n \geq 1} \left(\sup_{k \geq n} x_k \right)$

$= \inf_{n \geq 1} \{y_n\}$

$= \inf \{y_1, y_2, y_3, \dots\}$

$= \lim_{n \to \infty} y_n$ (as (y_n) is decreasing)

i.e $\boxed{\limsup x_n = \lim_{n \to \infty} \left[\sup_{k \geq n} x_k \right]}$

Let $z_n = \inf_{k \geq n} x_k = \inf \{x_n, x_{n+1}, x_{n+2}, \dots\}$

$\liminf x_n = \sup_{n \geq 1} \left[\inf_{k \geq n} x_k \right]$

$= \sup \{z_1, z_2, z_3, \dots\}$

$= \lim_{n \to \infty} z_n$ ((z_n) is increasing)

i.e $\boxed{\liminf x_n = \lim_{n \to \infty} \left[\inf_{k \geq n} x_k \right]}$

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Example - 28

Consider (x_n), where $x_n = (-1)^n + 1/n \quad \forall n \in \mathbb{N}$.

$$x_n = \begin{cases} 1 + 1/n & \text{if } n \text{ is even} \\ -1 + 1/n & \text{is } n \text{ is odd} \end{cases}$$

$$\lim \sup x_n = \lim_{n \to \infty} \left[\sup_{k \geq n} x_k \right]$$

$$= \lim_{n \to \infty} \left[\sup \{ x_n, x_{n+1}, x_{n+2}, \ldots \} \right]$$

$$= \lim_{n \to \infty} \left(1 + 1/n \right)$$

$$= \underline{\underline{1}}$$

$$\lim \inf x_n = \lim_{n \to \infty} \left[\inf_{k \geq n} x_k \right]$$

$$= \lim_{n \to \infty} \left(-1 + 1/n \right)$$

$$= -1$$

Example - 29

Consider (x_n), where $x_n = \frac{3 + 2(-1)^n}{5 + 3(-1)^n}$.

$$x_n = \begin{cases} 5/8 & \text{if } n \text{ is even} \\ 1/2 & \text{if } n \text{ is odd} \end{cases}$$

$$\limsup x_n = \lim_{n \to \infty} \left[\sup_{n \geq k} x_k \right]$$

$$= \lim_{n \to \infty} \left[\sup \{ x_n, x_{n+1}, x_{n+2}, \dots \} \right]$$

$$= \lim_{n \to \infty} 5/8$$

$$= 5/8$$

$$\liminf x_n = \lim_{n \to \infty} \left(\inf_{n \geq k} x_k \right)$$

$$= \lim_{n \to \infty} \left[\inf \{ x_n, x_{n+1}, x_{n+2}, \dots \} \right]$$

$$= \lim_{n \to \infty} \frac{1}{2}$$

$$= \frac{1}{2}$$

$$\boxed{\begin{array}{l} \lim_{n \to \infty} x_n = l \in \mathbb{R} \\ \quad \Leftrightarrow \limsup x_n = \liminf x_n = l \end{array}}$$

Example - 30

Consider (x_n), where $x_n = n(-1)^n \quad \forall\, n \in \mathbb{N}$

$$x_n = \begin{cases} n & \text{if } n \text{ is even} \\ -n & \text{if } n \text{ is odd} \end{cases}$$

<u>Note</u>

(x_n) is an unbounded Sequence.

$$\limsup x_n = \lim_{n \to \infty} \left(\sup (x_n, x_{n+1}, \dots) \right)$$

$$= \lim_{n \to \infty} (n)$$

$$= \infty$$

$$\liminf x_n = \lim_{n\to\infty}\left(\inf\left(x_n, x_{n+1}, x_{n+2}, \ldots\right)\right)$$

$$= \lim_{n\to\infty}(-n)$$

$$= -\infty$$

Theorem 12 (Root test)

Let $\sum a_n$ be a given series and let $\alpha = \limsup |a_n|^{1/n}$

a) If $\alpha < 1$, then $\sum a_n$ is absolutely Convergent.

b) If $\alpha > 1$, then $\sum a_n$ is divergent

c) If $\alpha = 1$, then test falls

Example - 31

Consider $\sum_{n=1}^{\infty} (-1)^{n+1}\left(\frac{2n+7}{8n-2}\right)^n$

Let $a_n = (-1)^{n+1}\left(\frac{2n+7}{8n-2}\right)^n$

$$\Rightarrow |a_n| = \left|(-1)^{n+1}\left(\frac{2n+7}{8n-2}\right)^n\right|$$

$$= \left(\frac{2n+7}{8n-2}\right)^n. \quad \text{Since } \frac{2n+7}{8n-2} > 0 \ \forall\ n \in \mathbb{N}$$

$$\Rightarrow |a_n|^{1/2} = \frac{2n+7}{8n-2}$$

$$\alpha = \limsup |a_n|^{1/n}$$

$$= \limsup \frac{2n+7}{8n-2} = \lim_{n\to\infty}\frac{2n+7}{8n-2}$$

$$= \lim_{n\to\infty}\left[\frac{2+7/n}{8-2/n}\right]$$

$$= \frac{2}{8} = \frac{1}{4} < 1$$

Then by the root test,

$\sum_{n=1}^{\infty} a_n$ is absolutely Convergent. Hence it is Convergent

Example - 32

Consider $\sum_{n=1}^{\infty} \left[\dfrac{3+(-1)^n}{3+2(-1)^n}\right]^n$

Let $a_n = \left[\dfrac{3+(-1)^n}{3+2(-1)^n}\right]^n$ for $n \in \mathbb{N}$

$\rightarrow$ $|a_n| = \left|\left(\dfrac{3+(-1)^n}{3+2(-1)^n}\right)^n\right| = \left(\dfrac{3+(-1)^n}{3+2(-1)^n}\right)^n$

$\alpha = \limsup |a_n|^{\frac{1}{n}}$

$= \limsup \left(\dfrac{3+(-1)^n}{3+2(-1)^n}\right)$ $\qquad a_n = \begin{cases} 4/5 & \text{if } n \text{ is even} \\ 2 & \text{if } n \text{ is odd.} \end{cases}$

$= \lim_{n\to\infty} 2$

$= 2 > 1$

$\therefore \sum a_n$ is divergent.

END.

power Series

A series of the form; $a_0 + a_1(x-c) + a_2(x-c)^2 + \dots + a_n(x-c) + \dots$

$a_0 + a_1(x-c) + a_2(x-c)^2 + \dots + a_n(x-c) + \dots$

$= \sum_{n=0}^{\infty} a_n (x-c)^n$ is called a power Series. centered at c, where $a_n \in \mathbb{R}$ for all $n \in \mathbb{N}$ and $c \in \mathbb{R}$.

Example - 01

Consider $\sum_{n=0}^{\infty} x^n$

$\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} 1 \times (x-0)^n$

Here $a_n = 1 \;\forall\, n \in \mathbb{N}$ and $c = 0$

$\therefore$ This is a power series centered at $c = 0$